Sanitation Performance Standards Compliance Guide

Introduction

On October 20, 1999, the Food Safety Inspection Service (FSIS) published a final rulemaking in the *Federal Register* that establishes regulatory sanitation performance standards applicable to all official meat and poultry establishments. (FSIS Docket 96-037F; 64 FR 56400) Performance standards set forth requirements in terms of an objective to be achieved, but do not prescribe the means to achieve that objective. Therefore, to meet the sanitation performance standards, establishments may develop and employ sanitation or processing procedures customized to the nature and volume of their production.

In this document, FSIS presents or references methods already proven to be effective in maintaining sanitary conditions in meat and poultry establishments. Past FSIS regulations and guidance, as well as recommendations from the 1999 Food Code and other technical sources, are included or cited. Establishments that follow the guidance in this document can be fairly certain that they are meeting the sanitation performance standards. Establishments should keep in mind, however, that each processing environment is unique and that in some cases, the methods presented in this document may be inadequate to ensure sanitary conditions or prevent the adulteration of meat and poultry products.

Establishments in compliance with past FSIS requirements may not want to change their procedures for maintaining sanitation. Such establishments may use this document as a reference, as it contains the past regulatory requirements governing sanitation (in the Appendix 1), as well as other guidance. Establishments that choose to innovate or customize their sanitation procedures also may find this document useful as a starting point for designing their new sanitation procedures.

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Format

Guidance for each performance standard is set forth as follows:

§ 416.2(b) Construction.

(1) Establishment buildings, including their structures,

The performance standard as stated in the regulations

Comments

In a recent FSIS Directive to its inspectors

Additional information concerning the performance standard

Food Code

6-201.11 Floors, Walls, and Ceilings.

Except as specified under § 6-201.14, the floors, floor

Relevant recommendations from the 1999 Food Code; we are including only sections that seem most applicable to most meat and poultry establishments; many establishments likely will find additional useful information in the Food Code that is not presented here

Other Sources of Guidance

The 1999 National Building Code published by BOCA

Applicable regulations of other Federal agencies and relevant recommendations from various technical sources

Finally, the specific sanitary practices described in this document **are not requirements**. Establishments must comply with the regulatory performance standards for sanitation cited below, but may do so by whatever means they

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determine to be appropriate. FSIS inspection personnel will verify that official establishments comply with the performance standards, regardless of whether the establishments follow the guidance in this document

Sources of Sanitation Information

The Food Code

The 1999 Food Code, published by the Food and Drug Administration, is a reference document for regulatory agencies responsible for overseeing food safety in retail outlets such as restaurants and grocery stores and institutions such as nursing homes and child care centers. It is neither federal law nor federal regulation and is not preemptive, but may be adopted and used by agencies at all levels of government that have responsibility for managing food safety risks at retail. Although the Food Code specifically addresses retail and institutional food service operations, many of its recommendations are applicable to official meat and poultry establishments.

The Food Code is available for free in several electronic formats (HTML, PDF, and Word Perfect) on the Internet. The Internet address is: http://vm.cfsan.fda.gov/~dms/foodcode.html <http://vm.cfsan.fda.gov/~dms/foodcode.html>

You also can purchase printed copies and CD-ROM and computer diskette versions of the 1999 Food Code from the National Technical Information Service (NTIS). You can order from NTIS on the Internet at: http://www.ntis.gov/index.html <http://www.ntis.gov/index.html> or by mail from:

U.S. Department of Commerce
Technology Administration
National Technical Information Service
5285 Port Royal Road, Springfield, VA 22161
(703) 605-6000, refer to report number PB99-115925

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Other Codes

In this guide, FSIS cites construction, plumbing, and sewage disposal guidance, standards, and codes developed other Federal agencies and by private standards organizations. FSIS does not require compliance with any of the private organizations' standards or codes and does not specifically endorse their use. However, these standards and codes provide useful information concerning construction, plumbing, and sewage disposal and, in many cases, compliance with them by meat and poultry establishments can ensure compliance with the sanitation performance standard regulations. Establishments, of course, may use other codes or information and should always comply with all applicable Federal, State, and local laws governing construction, plumbing, and sewage disposal. FSIS plans to reference additional codes and standards, as appropriate, in future versions of this compliance guide.

Comments on this Document

If the desired information cannot be found within the Compliance Guideline, FSIS recommends that users search the publicly posted Questions & Answers (Q&As) in the askFSIS knowledgebase or submit questions through askFSIS. Documenting these questions helps FSIS improve and refine present and future versions of the Compliance Guideline and associated issuances.

When submitting a question, use the Submit a Question tab, and enter the following information in the fields provided:

Subject Field: Sanitation Performance Standards Compliance Guide
Question Field: Enter question with as much detail as possible.
Product Field: Select **General Inspection Policy** from the drop-down menu.
Category Field: Select **Sanitation** from the drop-down menu.
Policy Arena: Select Domestic (U.S.) Only from the drop-down menu.

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The Performance Standards for Sanitation

§ 416.1 General rules.

Each official establishment must be operated and maintained in a manner sufficient to prevent the creation of insanitary conditions and to ensure that product is not adulterated.

Comments

Proper sanitation is a fundamental requirement under both the Federal Meat Inspection Act (FMIA) and the Poultry Products Inspection Act (PPIA). Meat and poultry products produced, packed, or held under insanitary conditions, where they may have become contaminated with filth or may have been rendered injurious to health, are adulterated.

§ 416.2 Establishment grounds and facilities.

Grounds and pest control. The grounds about an establishment must be maintained to prevent conditions that could lead to insanitary conditions, adulteration of product, or interfere with inspection by FSIS personnel. Establishments must have in place a pest management program to prevent the harborage and breeding of pests on the grounds and within establishment facilities. Pest control substances used must be safe and effective under the conditions of use and not be applied or stored in a manner that will result in the adulteration of product.

Comments

Grounds

Proper maintenance of the grounds about an establishment is essential for ensuring good sanitation. To keep vermin from breeding and to maintain sanitary conditions in general, an establishment should not

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allow trash to accumulate on its grounds, should store pallets and other equipment properly, and should keep its grounds drained. Actions an establishment needs to take will likely depend on the location of the establishment and the type of operations it conducts.

Under FSIS Directive 7640.1, "Inspection Duties Related to Facilities and Equipment, and Plant Operated Quality Control Programs," inspectors are directed to request from establishment management written designation of the official premises boundaries. Inspectors may use this information as reference when inspecting establishment grounds. Establishments should keep in mind, however, that they are responsible for preventing the adulteration of product even if the cause of the adulteration originates from conditions outside the designated boundaries of the establishment.

Pest Control

Meat and poultry establishments need to design and implement programs that ensure that product is not adulterated either by pests or by the products used to control them. Such a program should include not only the use of pesticides or other chemicals within or around an establishment, but also the maintenance of grounds to prevent harborage and breeding and measures to keep pests from entering establishment facilities. Pest control substances should be approved by EPA for use in food processing environments and be used in a manner that does not adulterate product or create insanitation. Under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), EPA reviews pesticide formulation, intended use, and other information; registers all pesticides for use in the United States; and prescribes labeling, use, and other regulatory requirements to prevent unreasonable adverse effects on the environment, including humans, wildlife, plants, and property.

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Any meat or poultry establishment using a pesticide must follow the FIFRA requirements. For more information on the use of chemicals, see § 416.4(c) and Appendix 2 below.

Food Code

6-102.11 Surface Characteristics.

A. The outdoor walking and driving areas shall be surfaced with concrete, asphalt, or gravel or other materials that have been effectively treated to minimize dust, facilitate maintenance, and prevent muddy conditions.
B. Exterior surfaces of buildings and mobile food establishments shall be of weather-resistant materials and shall comply with law.
C. Outdoor storage areas for refuse, recyclables, or returnables shall be of materials specified under § 5-501.11 and 5-501.12.

6-202.19 Outdoor Walking and Driving Surfaces, Graded to Drain.

Exterior walking and driving surfaces shall be graded to drain.

6-202.13 Insect Control Devices, Design and Installation.

A. Insect control devices that are used to electrocute or stun flying insects shall be designed to retain the insect within the device.
B. Insect control devices shall be installed so that:
    1. The devices are not located over a food preparation area; and
    2. Dead insects and insect fragments are prevented from being impelled onto or falling on exposed food; clean equipment, utensils, and linens; and unwrapped single-service and single-use articles.

7-202.12 Conditions of Use.

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Poisonous or toxic materials shall be:

A. Used according to:

1. Law and this Code,
2. Manufacturer's use directions included in labeling, and, for a pesticide, manufacturer's label instructions that state that use is allowed in a food establishment
3. The conditions of certification, if certification is required, for use of the pest control materials, and
4. Additional conditions that may be established by the regulatory authority; and

B. Applied so that:

1. A hazard to employees or other persons is not constituted, and
2. Contamination including toxic residues due to drip, drain, fog, splash or spray on food, equipment, utensils, linens, and single-service and single-use articles is prevented, and for a restricted-use pesticide, this is achieved by:
   a. Removing the items,
   b. Covering the items with impermeable covers, or
   c. Taking other appropriate preventive actions, and
   d. Cleaning and sanitizing equipment and utensils after the application.

C. A restricted use pesticide shall be applied only by an applicator certified as defined in 7 USC 136(e) Certified Applicator, of the Federal Insecticide, Fungicide and Rodenticide Act, or a person under the direct supervision of a certified applicator.

7-206.11 Restricted Use Pesticides, Criteria.

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Restricted use pesticides specified under § 7-202.12(C) shall meet the requirements specified in 40 CFR 152 Subpart I - Classification of Pesticides.

7-206.12 Rodent Bait Stations.

Rodent bait shall be contained in a covered, tamper-resistant bait station.

7-206.13 Tracking Powders, Pest Control and Monitoring.

A. A tracking powder pesticide may not be used in a food establishment.
B. If used, a nontoxic tracking powder such as talcum or flour may not contaminate food, equipment, utensils, linens, and single-service and single-use articles.

Other Sources of Guidance

§ 416.2(b) Construction.

a. Establishment buildings, including their structures, rooms, and compartments must be of sound construction, kept in good repair, and be of sufficient size to allow for processing, handling, and storage of product in a manner that does not result in product adulteration or the creation of insanitary conditions.
b. Walls, floors, and ceilings within establishments must be built of durable materials impervious to moisture and be cleaned and sanitized as necessary to prevent adulteration of product.
c. Walls, floors, ceilings, doors, windows, and other outside openings must be constructed and maintained to prevent the entrance of vermin, such as flies, rats, and mice.

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d. Rooms or compartments in which edible product is processed, handled, or stored must be separate and distinct from rooms or compartments in which inedible product is processed, handled, or stored, to the extent necessary to prevent product adulteration and the creation of insanitary conditions.

Comments

Here are some examples of noncompliance with these standards commonly reported by inspectors. They are not all inclusive; they represent noncompliance only with the performance standards in § 416.2(b), provisions (1) through (3).

- Doors not closing tightly allowing the entrance of vermin, dirt, or dust;
- Holes in ceilings or windows allowing the entrance of vermin, dirt, or dust;
- Scaling rust or scaling paint in edible areas on ceilings or walls;
- Walls in production area have mold growth.

(Section VI(B)(2))

In regard to § 416.2(b)(4), FSIS has allowed and will continue to allow edible and inedible products to be processed, handled, or stored in the same room or area, provided that measures are taken to prevent the adulteration of the edible product or insanitation that could lead to product adulteration. Typically, if an establishment processes, handles, or stores edible and inedible products in the same room or area, the products are separated by time and/or space and stringent controls are in place to prevent product adulteration.

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Food Code

6-201.11 Floors, Walls, and Ceilings.

Except as specified under § 6-201.14, the floors, floor coverings, walls, wall coverings, and ceilings shall be designed, constructed, and installed so they are smooth and easily cleanable, except that antislip floor coverings or applications may be used for safety reasons.

6-201.12 Floors, Walls, and Ceilings, Utility Lines.

A. Utility service lines and pipes may not be unnecessarily exposed.
B. Exposed utility service lines and pipes shall be installed so they do not obstruct or prevent cleaning of the floors, walls, or ceilings.
C. Exposed horizontal utility service lines and pipes may not be installed on the floor.

6-201.13 Floor and Wall Junctures, Covered, and Enclosed or Sealed.

A. In food establishments in which cleaning methods other than water flushing are used for cleaning floors, the floor and wall junctures shall be covered and closed to no larger than 1 mm (one thirty-second inch).
B. The floors in food establishments in which water flush cleaning methods are used shall be provided with drains and be graded to drain, and the floor and wall junctures shall be covered and sealed.

6-201.14 Floor Carpeting, Restrictions and Installation.

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A. A floor covering such as carpeting or similar material may not be installed as a floor covering in food preparation areas, walk-in refrigerators, warewashing areas, toilet room areas where handwashing lavatories, toilets, and urinals are located, refuse storage rooms, or other areas where the floor is subject to moisture, flushing, or spray cleaning methods.

B. If carpeting is installed as a floor covering in areas other than those specified under § (A) of this section, it shall be:

1. Securely attached to the floor with a durable mastic, by using a stretch and tack method, or by another method; and
2. Installed tightly against the wall under the coving or installed away from the wall with a space between the carpet and the wall and with the edges of the carpet secured by metal stripping or some other means.

6-201.15 Floor Covering, Mats and Duckboards.

Mats and duckboards shall be designed to be removable and easily cleanable.

6-201.16 Wall and Ceiling Coverings and Coatings.

A. Wall and ceiling covering materials shall be attached so that they are easily cleanable.

B. Except in areas used only for dry storage, concrete, porous blocks, or bricks used for indoor wall construction shall be finished and sealed to provide a smooth, nonabsorbent, easily cleanable surface.

6-201.17 Walls and Ceilings, Attachments.

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A. Except as specified in (B) of this section, attachments to walls and ceilings such as light fixtures, mechanical room ventilation system components, vent covers, wall mounted fans, decorative items, and other attachments shall be easily cleanable.

B. In a consumer area, wall and ceiling surfaces and decorative items and attachments that are provided for ambiance need not meet this requirement if they are kept clean.

6-201.18 Walls and Ceilings, Studs, Joists, and Rafters.

Studs, joists, and rafters may not be exposed in areas subject to moisture. This requirement does not apply to temporary food establishments.

6-202.15 Outer Openings, Protected.

A. Except as specified in (B) and under (C) of this section, outer openings of a food establishment shall be protected against the entry of insects and rodents by:

   1. Filling or closing holes and other gaps along floors, walls and ceilings;
   2. Closed, tight-fitting windows; and
   3. Solid self-closing, tight-fitting doors.

B. Paragraph (A) of this section does not apply if a food establishment opens into a larger structure, such as a mall, airport, or office building, or into an attached structure, such as a porch, and the outer openings from the larger or attached structure are protected against the entry of insects and rodents.

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C. Except as specified in (B) and (D) of this section, if the windows or doors of a food establishment, or of a larger structure within which a food establishment is located, are kept open for ventilation or other purposes or a temporary food establishment is not provided with windows and doors as specified under (A) of this section, the openings shall be protected against the entry of insects and rodents by:
    1. 16 mesh to 25.4mm (16 mesh to 1 inch) screens;
    2. Properly designed and installed air curtains; or
    3. Other effective means.
D. Paragraph (C) of this section does not apply if flying insects and other pests are absent due to the location of the establishment, the weather, or other limiting condition.

6-202.16 Exterior Walls and Roofs, Protective Barrier.

Perimeter walls and roofs of a food establishment shall effectively protect the establishment from the weather and the entry of insects, rodents, and other animals.

6-501.11 Repairing.

The physical facilities shall be maintained in good repair.

6-501.12 Cleaning, Frequency and Restrictions.

A. The physical facilities shall be cleaned as often as necessary to keep them clean.
B. Cleaning shall be done during periods when the least amount of food is exposed such as after closing. This requirement does not apply to cleaning that is necessary due to a spill or other accident.

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6-501.13 Cleaning Floors, Dustless Methods.

A. Except as specified in (B) of this section, only dustless methods of cleaning shall be used, such as wet cleaning, vacuum cleaning, mopping with treated dust mops, or sweeping using a broom and dust-arresting compounds.

B. Spills or drippage on floors that occur between normal floor cleaning times may be cleaned:

   1. Without the use of dust-arresting compounds; and
   2. In the case of liquid spills or drippage, with the use of a small amount of absorbent compound such as sawdust or diatomaceous earth applied immediately before spot cleaning.

Other Sources of Guidance

The 1999 National Building Code from the Building Officials and Code Administrators (BOCA) International, Inc., and the 1999 Standard Building Code from the Southern Building Code Congress International (SBCCI) both provide minimum standards for the construction of many types of buildings.

These codes may be obtained from:

Building Officials and Code Administrators International, Inc.
4051 W. Flossmoor Road, Country Club Hills, Illinois 60478-5795 USA.
Telephone (main): (708) 799-2300
Telephone (publications orders): 1-800-214-4321 ext.777
FAX (publications orders): 1-800-214-7167
Email: codes@bocai.org
Internet home page: http://www.bocai.org <http://www.bocai.org/>

and:

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Southern Building Code Congress International
900 Montclair Road, Birmingham, AL 35213-1206
Telephone: (205) 591-1853
FAX: (205) 591-0775
Email: info@sbcci.org
Internet home page: http://www.sbcci.org <http://www.sbcci.org/>

§ 416.2(c) Light.

Lighting of good quality and sufficient intensity to ensure that sanitary conditions are maintained and that product is not adulterated must be provided in areas where food is processed, handled, stored, or examined; where equipment and utensils are cleaned; and in hand-washing areas, dressing and locker rooms, and toilets.

Comments

Establishments should keep in mind that their lighting should be sufficient not only to allow their own employees to maintain sanitation and prevent product adulteration, but also to allow FSIS inspection personnel to verify that conditions are sanitary and product is not adulterated. This does not mean, however, that lighting sufficiency is to be determined subjectively, by the inspector. Establishments must determine which intensities and qualities of light are appropriate in different processing environments. FSIS will direct its inspection personnel to make judgments accordingly.

Establishments also should keep in mind the specific lighting intensity requirements for inspector and reprocessing stations, in § 307.2 and 381.36 of the regulations, are still in effect.

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Food Code

6-303.11 Intensity.

The light intensity shall be:

- At least 110 lux (10 foot candles) at a distance of 75 cm (30 inches) above the floor, in walk-in refrigeration units and dry food storage areas and in other areas and rooms during periods of cleaning;
- At least 220 lux (20 foot candles):
  1. At a surface where food is provided for consumer self-service such as buffets and salad bars or where fresh produce or packaged foods are sold or offered for consumption;
  2. Inside equipment such as reach-in and under-counter refrigerators;
  3. At a distance of 75 cm (30 inches) above the floor in areas used for handwashing, warewashing, and equipment and utensil storage, and in toilet rooms; and
- At least 540 lux (50 foot candles) at a surface where a food employee is working with food or working with utensils or equipment such as knives, slicers, grinders, or saws where employee safety is a factor.

Other Sources of Guidance

§ 416.2(d) Ventilation.

Ventilation adequate to control odors, vapors, and condensation to the extent necessary to prevent adulteration of product and the creation of insanitary conditions must be provided.

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Comments

In regard to condensation, keep in mind that some forms are unavoidable and acceptable within a food processing environment, since they will neither adulterate product nor create insanitary conditions. Other forms of condensation are expected, but must be controlled by the establishment, and others are unacceptable at any time. Examples of different types of condensation and corresponding FSIS responses (from FSIS Notice 31-98) follow:

SITUATIONS INVOLVING CONDENSATION IN WHICH NO ACTION IS REQUIRED

In certain situations, condensation within an official establishment has no affect on product safety, sanitary conditions, or inspection. If inspection program personnel determine that such a situation exists, no action is necessary by him/her or the establishment. Some examples follow:

1. Condensation forms on the underside of a stainless steel vessel lid during cooking.
2. The packaging of packaged entrees or soups comes into contact with condensation which has formed as a result of freezing operations.
3. Condensation forms on the wall or ceiling of a loading dock where canned products are stored in wrapped boxes on palettes. (Although this situation may not threaten product safety or impede inspection, establishments should avoid allowing excessive condensation to form anywhere food is processed or stored.)

SITUATIONS IN WHICH CONDENSATION IS EXPECTED AND CONTROLLED BY THE OFFICIAL ESTABLISHMENT

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In other situations, establishments expect condensation to form as a result of certain operations and take action to ensure that the condensation does not adulterate product or create insanitary conditions. Such actions must be documented in the establishment's Sanitation Standard Operating Procedures (Sanitation SOP's). Most often, establishments will control such condensation by cleaning and sanitizing, on a daily or as-needed basis, the surface(s) where the condensation is expected to form. Examples of such surfaces include:

1. The inside or outside of stainless production chutes.
2. Ceilings over open kettle cooking areas and over poultry chill vats.
3. The outside of stainless steel ice vats or ice chutes in chill areas.

SITUATIONS IN WHICH INSPECTION PROGRAM PERSONNEL MUST TAKE ACTION

In some situations, condensation clearly adulterates product, creates insanitary conditions, and/or interferes with inspection. Some examples follow:

1. Heavily beaded condensation forms on a ceiling or wall of a processing area that is not regularly cleaned and sanitized in accordance with the establishment's SSOP's (an insanitary condition is created that could lead to the adulteration of product).
2. Condensate from a cooler ceiling drips onto carcasses.
3. Condensate from refrigeration unit surfaces, which have not been cleaned and sanitized, drips onto exposed product.
4. Condensate from a loading dock ceiling or wall drips onto boxes of boneless beef, breaking down the packaging.

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Food Code

4-301.14 Ventilation Hood Systems, Adequacy.

Ventilation hood systems and devices shall be sufficient in number and capacity to prevent grease or condensation from collecting on walls and ceilings.

6-202.12 Heating, Ventilating, Air Conditioning System Vents.

Heating, ventilating, and air conditioning systems shall be designed and installed so that make-up air intake and exhaust vents do not cause contamination of food, food-contact surfaces, equipment, or utensils.

6-304.11 Mechanical.

If necessary to keep rooms free of excessive heat, steam, condensation, vapors, obnoxious odors, smoke, and fumes, mechanical ventilation of sufficient capacity shall be provided.

Other Sources of Guidance

§ 416.2(e) Plumbing.

Plumbing systems must be installed and maintained to:

a. Carry sufficient quantities of water to required locations throughout the establishment;
b. Properly convey sewage and liquid disposable waste from the establishment;
c. Prevent adulteration of product, water supplies, equipment, or utensils, and maintain sanitary conditions throughout the establishment;

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d. Provide adequate floor drainage in all areas where floors are subject to flooding-type cleaning or where normal operations release or discharge water or other liquid waste on the floor;

e. Prevent back-flow conditions in and cross-connection between piping systems that discharge waste water or sewage and piping systems that carry water for product manufacturing; and

f. Prevent the backup of sewer gases.

Comments

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Food Code

5-201.11 Approved.

- A plumbing system and hoses conveying water shall be constructed and repaired with approved materials according to law.
- A water filter shall be made of safe materials.

5-202.11 Approved System and Cleanable Fixtures.

- A plumbing system shall be designed, constructed, and installed according to law.
- A plumbing fixture such as a handwashing lavatory, toilet, or urinal shall be easily cleanable.

5-202.13 Backflow Prevention, Air Gap.

An air gap between the water supply inlet and the flood level rim of the plumbing fixture, equipment, or nonfood equipment shall be at least twice the diameter of the water supply inlet and may not be less than 25 mm (1 inch).

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5-202.14 Backflow Prevention Device, Design Standard.

A backflow or backsiphonage prevention device installed on a water supply system shall meet American Society of Sanitary Engineering (A.S.S.E.) standards for construction, installation, maintenance, inspection, and testing for that specific application and type of device.

5-202.15 Conditioning Device, Design.

A water filter, screen, and other water conditioning device installed on water lines shall be designed to facilitate disassembly for periodic servicing and cleaning. A water filter element shall be of the replaceable type.

Other Sources of Guidance

The 1997 International Plumbing Code from BOCA, 1997 Standard Plumbing Code from SBCCI, and the 1997 Uniform Plumbing Code from the International Association of Plumbing and Mechanical Officials (IAPMO) provide basic principals and minimum requirements for plumbing systems. Notably, these codes address some of the issues relevant to plumbing in food processing facilities. For contact information for BOCA and SBCCI, see "Other Sources of Guidance" under "Construction" above. To obtain materials from IAPMO, contact:

International Association of Plumbing and Mechanical Officials
20001 E Walnut Drive South
Walnut CA 91789-2825
Telephone: (909) 595-8449
FAX (publications orders): (909) 598-4720
Internet home page: http://www.iapmo.org/iapmo/iapmo.html

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Also, The American Society of Mechanical Engineers (ASME) has published guidance regarding the "design, quality and performance of serving check backwater valves for use in building drainage systems." The document, entitled "Backwater Valves," is available from:

ASME International
Three Park Avenue
New York, NY 10016-5990
Phone: 1-800-843-2763
Fax: 1-973-882-1717
Email: infocentral@asme.org
Internet home page: http://www.asme.org

Finally, the Environmental Protection Agency (EPA) Office of Ground Water and Drinking Water (OGWDW) has published a document concerning the protection of potable water from contamination resulting from cross-connection. The document, entitled "Cross-Connection Control Manual," is available from OGWDW at the following address:

Office of Ground Water and Drinking Water (4601)
401 M Street, SW
Washington, DC 20460-0003
Phone: 202-260-5543
Internet home page: http://www.epa.gov/safewater/about.html#fax

§ 416.2(f) Sewage disposal.

Sewage must be disposed into a sewage system separate from all other drainage lines or disposed of through other means sufficient to prevent backup of sewage into areas where product is processed, handled, or

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stored. When the sewage disposal system is a private system requiring approval by a State or local health authority, the establishment must furnish FSIS with the letter of approval from that authority upon request.

Comments

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Food Code

4-204.17 Ice Units, Separation of Drains.

Liquid waste drain lines may not pass through an ice machine or ice storage bin.

5-402.10 Establishment Drainage System.

Food establishment drainage systems, including grease traps, that convey sewage shall be designed and installed as specified under § 5-202.11(A).

5-402.11 Backflow Prevention.

A. Except as specified in (B) and (C) of this section, a direct connection may not exist between the sewage system and a drain originating from equipment in which food, portable equipment, or utensils are placed.
B. If allowed by law, a warewashing machine may have a direct connection between its waste outlet and a floor drain when the machine is located within 1.5 m (5 feet) of a trapped floor drain and the machine outlet is connected to the inlet side of a properly vented floor drain trap.
C. If allowed by law, a warewashing or culinary sink may have a direct connection.

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5-402.12 Grease Trap.

If used, a grease trap shall be located to be easily accessible for cleaning.

5-402.13 Conveying Sewage.

Sewage shall be conveyed to the point of disposal through an approved sanitary sewage system or other system, including use of sewage transport vehicles, waste retention tanks, pumps, pipes, hoses, and connections that are constructed, maintained, and operated according to law.

5-402.15 Flushing a Waste Retention Tank.

A tank for liquid waste retention shall be thoroughly flushed and drained in a sanitary manner during the servicing operation.

5-403.11 Approved Sewage Disposal System.

Sewage shall be disposed through an approved facility that is:

A. A public sewage treatment plant; or
B. An individual sewage disposal system that is sized, constructed, maintained, and operated according to law.

5-403.12 Other Liquid Wastes and Rainwater.

Condensate drainage and other nonsewage liquids and rainwater shall be drained from point of discharge to disposal according to law.

Other Sources of Guidance

The plumbing codes listed above under "Other Sources of Guidance" for § 416.2(e) contain much information relative the proper disposal of sewage.

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§ 416.2(g)Water supply and water, ice, and solution reuse.

a. A supply of running water that complies with the National Primary Drinking Water regulations (40 CFR Part 141), at a suitable temperature and under pressure as needed, must be provided in all areas where required (for processing product, for cleaning rooms and equipment, utensils, and packaging materials, for employee sanitary facilities, etc.). If an establishment uses a municipal water supply, it must make available to FSIS, upon request, a water report, issued under the authority of the State or local health agency, certifying or attesting to the potability of the water supply. If an establishment uses a private well for its water supply, it must make available to FSIS, upon request, documentation certifying the potability of the water supply, that has been renewed at least semi-annually.
b. Water, ice, and solutions (such as brine, liquid smoke, or propylene glycol) used to chill or cook ready-to-eat product may be reused for the same purpose, provided that they are maintained free of pathogenic organisms and fecal coliform organisms and that other physical, chemical, and microbiological contamination have been reduced to prevent adulteration of product.
c. Water, ice, and solutions used to chill or wash raw product may be reused for the same purpose provided that measures are taken to reduce physical, chemical, and microbiological contamination so as to prevent contamination or adulteration of product. Reuse water which has come into contact with raw product may not be used on ready-to-eat product.

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d. Reconditioned water that has never contained human waste and that has been treated by an onsite advanced wastewater treatment facility may be used on raw product, except in product formulation, and throughout the facility in edible and inedible production areas, provided that measures are taken to ensure that this water meets the criteria prescribed in paragraph (g)(1) of this section. Product, facilities, equipment, and utensils coming in contact with this water must undergo a separate final rinse with non-reconditioned water that meets the criteria prescribed in paragraph (g)(1) of this section.

e. Any water that has never contained human waste and that is free of pathogenic organisms may be used in edible and inedible product areas, provided it does not contact edible product. For example, such reuse water may be used to move heavy solids, flush the bottom of open evisceration troughs, or to wash antemortem areas, livestock pens, trucks, poultry cages, picker aprons, picking room floors, and similar areas within the establishment.

f. Water that does not meet the use conditions of paragraphs (g)(1) through (g)(5) of this section may not be used in areas where edible product is handled or prepared or in any manner that would allow it to adulterate edible product or create insanitary conditions.

Comments

FSIS has developed the following guidance for water, ice and solution reuse.

ICE REUSE

Ice from ice packed poultry may be reused to repack raw whole birds or parts. The following are recommended:

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- Establish a procedure to assure that ice is collected and held in a container that drains freely and in a sanitary manner. The procedure should address collection and washing of ice before it is reused.
- Establish a procedure for identifying reused ice from fresh ice.
- The ice or the product should be packaged in an impervious, sealed container, such as a plastic bag, to prevent direct contact between the product and ice.
- Ice used on raw product should not be reused on any heat processed partially- or fully-cooked product.
- The ice should be free of any observable foreign material as well as large particles of poultry meat and fat. If the ice is washed, continuous drainage should be maintained during the washing procedure.
- Ice from damaged containers should not be used.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

BRINE REUSE

Brine may be reused d to chill cooked product for various lengths of time based on the type of casing, salinity, and temperature.

Brine solution that is reused to chill raw or heat -treated, raw but not fully cooked product (example- e.g., smoked bacon) should be reconditioned in a manner to prevent the brine solution from becoming contaminated and adulterating the product.

Brine reuse to chill raw product should follow the same criteria as brine reused to chill heat-treated, not fully cooked product.

The following are recommended:

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- Establish procedures for monitoring the temperature, salinity, and free chlorine concentration of the brine being reused to chill heat-treated product.
- Establish an ongoing microbiological plan to ensure that the brine solution is maintained pathogen free. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reused brine solution:

Cooked Product

Analysis	Frequency	Action Level
Total Plate Count	Daily	> 2500 cfu/ml
Total Coliform	Weekly	Positive
Fecal Coliform	Weekly	Positive

Raw or Heat-Treated, Raw Not Fully Cooked Product (i.e. Bacon Bellies)

Analysis	Frequency	Action Level
Total Plate Count	Daily	> 5000 cfu/ml
Total Coliform	Weekly	>10 cfu/ml

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Fecal Coliform	Weekly	Positive

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Visible contamination defects should be removed from the product before it is placed in the brine solution.
- The solution should be kept free of visible meat and fat particles and other objectionable conditions by effective methods such as filtration, skimming, or overflow.

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- When the brine solution is used without reconditioning for one shift or longer, the solution should be discarded at the following specified intervals, and all equipment, tanks, lines should be thoroughly cleaned and sanitized:

Heat-Treated Product		
Duration of Use	Classes	Additional Conditions
One production shift	All Classes: No casing Perforated casing Edible casing Semipermeable casing Impermeable casing	None
Up to 24 hours	All classes: No casing Perforated casing Edible casing Semipermeable casing Impermeable casing	1. Minimum salt 5% (19' salimeter) 2. Maintain 40' F. or lower

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Up to 1 week	One class: Semipermeable casings Impermeable casing	1. Minimum salt 9% (32' salimeter) 2. Maintain 28' F. or lower
Up to 4 weeks	One class: Semipermeable casing Impermeable casing	1. Minimum salt 20% (76' salimeter) 2. Maintain 10' F. or lower

- Cooked product, for example frankfurters, cannot be chilled in a brine solution that has been used to chill raw or heat-treated, not fully cooked product, for example, bacon bellies. (Raw product may be chilled after cooked product)..
- Products with semipermeable or impermeable casing that are being chilled in brine that is being reused for longer than 24 hours should be trimmed if they have broken casings or have been similarly exposed. The trimmings should be discarded as inedible.
- A free chlorine concentration of 1-5 ppm should be maintained in the reuse brine solution.

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- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

COOK AND CHILL WATER REUSE

Water may be reused to cook product and to chill cooked product. The following are recommended:

- Establish procedures for monitoring the temperature of the cook or chill water, and free chlorine concentration of the chill water being reused to chill cooked product.

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- Establish an ongoing microbiological plan to ensure that the continuous safety of reuse cook or chill water is maintained pathogen free. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reused cook and/or chill water:

Chill Water		
Analysis	Frequency	Action Level
Total Plate Count	Daily	>500 cfu/ml
Total Coliform	Weekly	Positive
Fecal Coliform	Weekly	Positive
Turbidity	Weekly	>5 NTU
Cook Water		
Analysis	Frequency	Action Level
Total Plate Count	Daily	>500 cfu/ml
Gas Forming Anaerobes	Weekly	Positive
Total Coliform	Weekly	Positive

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Turbidity	Weekly	>5 NTU

- Initially, frequency of microbial testing would be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Visible contamination defects should be removed from the product before it is placed in the cook and/or chill water.
- The cook or chill water should be kept free of visible meat and fat particles and other objectionable conditions by effective methods such as filtration, skimming, or overflow.
- The chill water should be maintained at a temperature of 50' F. or less.
- The cook water should be maintained at a temperature of 150' F. or higher.
- A free chlorine concentration of 1-5 ppm should be maintained in the reuse chill water
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

PROPYLENE GLYCOL REUSE

Propylene glycol solution may be reused to chill raw product such as hamburger chubs, sausage chubs, and bagged poultry for up to an indefinite length of time. The following are recommended:

- Establish procedures for monitoring the temperature, propylene glycol concentration, and free chlorine concentration of the propylene glycol solution being reused to chill raw product.

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- Establish an ongoing microbiological plan to ensure the continuous safety of the propylene glycol solution. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reused propylene glycol solution:

Analysis	Frequency	Action Level
Total Plate Count	Weekly	>500 cfu/ml
Total Coliform	Weekly	>10 cfu/ml
Fecal Coliform	Weekly	Positive

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Visible contamination defects should be removed from the product before it is placed in the propylene glycol solution.
- The propylene glycol solution should be kept free of visible meat and fat particles and other objectionable conditions by effective methods such as filtration, skimming, or overflow.
- The propylene glycol solution should be maintained at a temperature of 10' F. or less during production hours and 40' F. or less during nonproduction hours.

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- The propylene glycol should be of a type that is authorized for use for immersion freezing of meat and poultry products.
- The product should be enclosed in a package that does not allow the propylene glycol solution directly or indirectly to contact it. It is recommended that product be enclosed within an impervious package.
- Products that are exposed to the propylene glycol solution should be appropriately handled as contaminated product. One appropriate way of handling the contaminated product would be to rewash the product by water spraying. All traces of refrigerant should be removed before product is passed for food. If all contamination cannot be removed by water washing or washing or trimming cannot remove all contamination, the affected portion should be condemned.
- The propylene glycol solution should be adequately removed from the packaged products after freezing and before placing into shipping containers by effective methods such as water spray washing equipment.
- A free chlorine concentration of 1-5 ppm is recommended to be maintained in the propylene glycol.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

CHILLER OVERFLOW WATER REUSE

Overflow water from the poultry chilling units may be reused to move heavy solids in eviscerating troughs (not to flush sides of trough), scald tank, feather flow-aways, picker aprons, and washing picking room floors. The following are recommended:

- Establish a procedure to assure that chiller overflow water is collected and used in a sanitary manner.

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- Chiller overflow water added to the scalder should be a minimum of 140' F.
- The use of chiller overflow water to rinse picker aprons and wash picking room floors should be used in a manner that prevents cross-contamination to other areas of the plant such as from that due to employee traffic.
- The chiller overflow water should be kept free of visible solids.
- The chiller overflow water is collected and handled in a sanitary manner.
- Establish an ongoing microbiological plan to ensure that the continuous safety of the chiller overflow reuse water is maintained pathogen free. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reused chiller overflow water:

Analysis	**Frequency**	**Action Level**
Total Coliforms	Weekly	Positive
Fecal Coliforms	Weekly	Positive
Salmonella	Weekly	Positive
Staphylococcus aureus (coagulase positive staphylococci)	Weekly	Positive

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- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

CONDENSER OR COMPRESSOR WATER REUSE

Water from condensers or compressors may be reused in edible and inedible product areas providing that it is maintained pathogen free. The following are recommended:

- The reuse condenser or compressor water should be collected and handled in a sanitary manner.
- The reuse condenser or compressor water should be maintained in a manner that prevents the solution from becoming contaminated such as with coliforms, oil and grease, refrigerant, or heavy metals that can adulterate product.adulterate product.
- A free chlorine concentration of 1-5 ppm should be maintained in the reuse condenser or compressor water.

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- An ongoing monitoring plan should be established to ensure that the continuous safety of the reuse condenser and compressor water is maintained pathogen free. The monitoring plan should cover the type and frequency of any physical, chemical, and microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reuse chill water:

Analysis	**Frequency**	**Action Level**
Total Plate Count	Weekly	>500 cfu/ml
Total Coliform	Weekly	Positive
Fecal Coliform	Weekly	Positive
Turbidity	Weekly	no samples > 5 NTU

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.

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- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

REUSE WATER TO FLUME CHICKEN FEET (PAWS)

Poultry chiller overflow water and water used to flume chicken feet (paws) may be used to flume chicken feet including through an in-line paw chiller. The following are recommended:

- Potable water should be added periodically to prevent organic matter buildup.
- The chiller overflow water and paw flume water should be kept free of visible solids.
- A free chlorine concentration of 1-5 ppm should be maintained in the reuse water used to convey chicken feet (paws).
- An ongoing microbiological monitoring plan should be established to ensure and ensure the continuous safety of that the reuse chiller overflow water and paw flume water used to flume chicken paws is maintained pathogen free. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reuse chiller overflow water and paw flume water:

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Analysis	Frequency	Action Level
Total Coliforms	Weekly	Positive
Fecal Coliforms	Weekly	Positive
Salmonella	Weekly	Positive
Staphylococcus aureus (coagulase positive staphylococci)	Weekly	Positive

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

REUSE WATER TO BE USED TO WASH LIVESTOCK PENS, TRUCKS, POULTRY CAGES, AND SIMILAR AREAS

Water from establishment's secondary and tertiary wastewater treatment facility or other processing water may be reused to wash livestock pens, trucks, poultry cages, and other similar areas. The following are recommended:

- Water from the establishment's wastewater treatment facility or other processing water to be used for washing should be kept free of visible solids.

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- A free chlorine concentration of 1-5 ppm should be maintained in the reuse water.
- The water from the establishment's wastewater treatment facility or other processing water should be collected and handled in a sanitary manner.
- The establishment's wastewater treatment system must not be treating human waste. Human waste must be kept separate from plant waste and not be commingled at the wastewater treatment system.

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- An ongoing microbiological monitoring plan should be established to ensure that the continuous safety of the reuse water from the establishment's wastewater treatment facility or other processing water are maintained pathogen free. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reuse water from the establishment's wastewater treatment facility or other processing water:

Analysis	**Frequency**	**Action Level**
Total Coliforms	Weekly	Positive
Fecal Coliforms	Weekly	Positive
Salmonella	Weekly	Positive
Staphylococcus aureus (coagulase positive staphylococci)	Weekly	Positive

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing until system controls are re-established.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

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REUSE WATER TO BE USED TO WASH INEDIBLE PRODUCT AREAS

Water from throughout the plant may be reused in inedible product areas (i.e. washing offal sump screen, flushing feather flow-away troughs, flushing eviscerating troughs that are covered with metal plates, etc.). The following are recommended:

- The reuse water should be used in a manner that prevents cross-contamination to other areas of the plant such as from that due to employee traffic.
- The reuse water to be used should not violate any OSHA requirements.
- The reuse water to be used in inedible areas under FDA jurisdiction, such as pet food areas, must also meet FDA requirements.
- The reuse water should be kept free of visible solids.
- The reuse water is collected and handled in a sanitary manner.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

REUSE WATER FROM AN ADVANCED WASTEWATER TREATMENT FACILITY

Reuse water from an advanced wastewater treatment facility may be used on edible product (but not in product formulation) and throughout the plant in edible and inedible production areas. The following are recommended:

- An advanced wastewater treatment facility should meet EPA requirements.

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- The establishment's advanced wastewater treatment must not be treating human waste. Human waste must be kept separate from plant waste and not be commingled at the advanced wastewater treatment facility.
- The establishment should have qualified and trained personnel who monitor, regulate, and record the wastewater treatment system.
- The establishment should have a program in place that identifies, monitors, and records the treatment measures necessary for safe and effective operation of the wastewater treatment facility.
- The potable and reuse water lines should be identified and separated except at junctions where appropriate valves, etc. protect the potable water supply.
- Dual check valves, alarms, etc., should be in place in case the reuse water system malfunctions to prevent the reuse water from contaminating the potable water supply.
- A "Fail-Safe" system should be in place to prevent substandard reuse water from entering the "end use" part of the system and contaminating edible product.* The reuse water should be tested for heavy metals at least once a year and meet the appropriate EPA's Maximum Contaminant Levels (MCLs).
- A final potable water rinse should be applied to any edible product and any equipment that contacts reuse water.
- The "End Use" reused water should be monitored and tested daily to ensure that the reuse water meets the criteria for the intended use.

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- The reuse water should meet the following "Safe for the Intended Use" EPA Criteria:
    1. Microbiological analysis
        1. Total Aerobic Plate Count <= 500 CFU/M
        2. Total Coliforms - None
        3. E. coli - None

- Chemical analysis
- Total Organic Carbon (TOC) <= 100 MG/L
- Physical analysis
- Turbidity - <= 5% of samples analyzed >= 1 NTU by EPA nephelometry method or equivalent method; no samples > 5 NTU
- The reuse water should be tested for heavy metals at least once a year and meet the appropriate EPA Maximum Contaminant Levels (MCL's).
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

REUSE WATER IN VAPOR LINES FROM DEODORIZERS

Water in vapor lines from deodorizers (condensers) used in preparation of lard and similar edible product may be reused for the same identical use. The following are recommended:

- The complete drainage and disposal of the reused water, effective cleaning of the equipment, and renewal with fresh potable water should be accomplished often enough to assure an acceptable supply of reuse water for the preparation of lard and similar edible product.
- The reuse water in vapor lines from deodorizers should be maintained in a manner that prevents the solution from becoming contaminated such as with coliforms, oil, or grease that can adulterate the product.

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- An ongoing monitoring plan should be established to ensure that the continuous safety of the reuse water in vapor lines from deodorizers is maintained pathogen free. The monitoring plan should cover the type and frequency of any physical, chemical, and microbiological analysis, and action limits (upper/lower control limits), and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reuse water:

Analysis	Frequency	Action Level
Total Plate Count	Weekly	>500 cfu/ml
Total Coliform	Weekly	Positive
Fecal Coliform	Weekly	Positive
Turbidity	Weekly	No samples > 5 NTU

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Establish procedures to correct deficiencies that occur to prevent reoccurrence.

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REUSE WATER FROM SINGLE OR MULTIPLE POINT SOURCES CAN BE USED FOR SINGLE OR MULTIPLE POINT SOURCES IN THE SLAUGHTER PROCESS

Reuse water from any slaughter process location(s) (e.g., scalder, inside/outside bird washer, chiller overflow water, etc) can be used at any location(s) in the slaughter process including for the chiller make-up water and for general sanitation purposes. For example, chiller overflow water and water from the final bird washer that are reconditioned and meet the criteria listed below can be reused in the scalder, throughout the eviscerating line, inside/outside bird washer, final bird washer, chiller make-up water and for general sanitation purposes. Consequently, since the reuse water is reconditioned to a higher standard since it can replace potable water used during the slaughter process, it needs to meet a higher water reuse standard than pathogen free. The following are recommended:

- Establish procedures for monitoring turbidity and concentration of the water being reused during the slaughter process.
- A free chlorine concentration of 1-5 ppm should be maintained in the reuse water.
- The potable and reuse water lines should be identified and separated except at junctions where appropriate valves, etc, protect the potable water supply.
- A system should be in place to prevent substandard reuse water from entering the "end use" part of the system and contaminate edible product.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

- Establish an ongoing microbiological plan to ensure the continuous safety of reuse water during the slaughter process. The monitoring plan should cover the type and frequency of any microbiological analysis, and action limits (upper/lower control limits) and actions taken to ensure product safety when those limits are exceeded. It is recommended that the establishment perform the following ongoing monitoring of the reconditioned water:

Analysis	**Frequency**	**Action level**
Total Plate Count	Weekly	>500 cfu/ml
Total Coliform	Weekly	Positive
Fecal Coliform	Weekly	Positive
Turbidity	Daily	>5 NTU

- Initially, frequency of microbial testing should be at the highest level until control is established. Reduced testing may be appropriate once control has been established. However, loss of control may necessitate a return to increased testing frequency until system controls are re-established.
- Establish procedures to correct deficiencies that occur and to prevent reoccurrence.

Food Code

5-101.11 Approved System.

Drinking water shall be obtained from an approved source that is:

1. A public water system; or

Guidance documents lack the force and effect of law, unless expressly authorized by statute or incorporated into a contract.
USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

2. A nonpublic water system that is constructed, maintained, and operated according to law.

5-101.12 System Flushing and Disinfection.

A drinking water system shall be flushed and disinfected before being placed in service after construction, repair, or modification and after an emergency situation, such as a flood, that may introduce contaminants to the system.

5-102.11 Standards.

Except as specified under § 5-102.12:

1. Water from a public water system shall meet 40 CFR 141 - National Primary Drinking Water Regulations and state drinking water quality standards; and
2. Water from a nonpublic water system shall meet state drinking water quality standards.

5-102.12 Nondrinking Water.

1. Nondrinking water shall be used only for nonculinary purposes such as air conditioning, nonfood equipment cooling, fire protection, and irrigation.

5-102.13 Sampling.

Except when used as specified under § 5-102.12, water from a nonpublic water system shall be sampled and tested at least annually and as required by state water quality regulations.

5-102.14 Sample Report.

Guidance documents lack the force and effect of law, unless expressly authorized by statute or incorporated into a contract.
USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

The most recent sample report for the nonpublic water system shall be retained on file in the food establishment or the report shall be maintained as specified by state water quality regulations.

Other Sources of Guidance

§ 416.2(h) Dressing rooms, lavatories, and toilets.

Dressing rooms, toilet rooms, and urinals must be sufficient in number, ample in size, conveniently located, and maintained in a sanitary condition and in good repair at all times to ensure cleanliness of all persons handling any product. They must be separate from the rooms and compartments in which products are processed, stored, or handled.

Lavatories with running hot and cold water, soap, and towels, must be placed in or near toilet and urinal rooms and at such other places in the establishment as necessary to ensure cleanliness of all persons handling any product.

Refuse receptacles must be constructed and maintained in a manner that protects against the creation of insanitary conditions and the adulteration of product.

Comments

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Food Code

5-202.12 Handwashing Lavatory, Water Temperature, and Flow.

1. A handwashing lavatory shall be equipped to provide water at a temperature of at least 43°C (110°F) through a mixing valve or combination faucet.
2. A steam mixing valve may not be used at a handwashing lavatory.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

3. A self-closing, slow-closing, or metering faucet shall provide a flow of water for at least 15 seconds without the need to reactivate the faucet.

5-501.10 Indoor Storage Area.

If located within the food establishment, a storage area for refuse, recyclables, and returnables shall meet the requirements specified under 6-101.11, 6-201.11 - 6-201.18, 6-202.15, and 6-202.16.

5-501.13 Receptacles.

Except as specified in (B) of this section, receptacles and waste handling units for refuse, recyclables, and returnables and for use with materials containing food residue shall be durable, cleanable, insect- and rodent-resistant, leakproof, and nonabsorbent.
Plastic bags and wet strength paper bags may be used to line receptacles for storage inside the food establishment, or within closed outside receptacles.

Other Sources of Guidance

The Occupational Safety and Health Administration of the Department of Labor has promulgated regulations concerning toilet facilities in the workplace in 29 CFR 1910.141, "Sanitation." Paragraph (c)(1)(i) sets forth requirements for the number of toilet facilities in all permanent places of employment. Official meat and poultry establishments are governed by these requirements:

1910.141 (c)(1)(i)

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Except as otherwise indicated in this paragraph (c)(1)(i), toilet facilities, in toilet rooms separate for each sex, shall be provided in all places of employment in accordance with table J-1 of this section. The number of facilities to be provided for each sex shall be based on the number of employees of that sex for whom the facilities are furnished. Where toilet rooms will be occupied by no more than one person at a time, can be locked from the inside, and contain at least one water closet, separate toilet rooms for each sex need not be provided. Where such single-occupancy rooms have more than one toilet facility, only one such facility in each toilet room shall be counted for the purpose of table J-1.

TABLE J-1

Number of employees	Minimum number of water Closets (1)
1 to 15	1
16 to 35	2
36 to 55	3
56 to 80	4
81 to 110	5
111 to 150	6
Over 150	(2)

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Footnote(1) Where toilet facilities will not be used by women, urinals may be provided instead of water closets, except that the number of water closets in such cases shall not be reduced to less than 2/3 of the minimum specified.

Footnote(2) 1 additional fixture for each additional 40 employees.

§ 416.3 Equipment and utensils.

a. Equipment and utensils used for processing or otherwise handling edible product or ingredients must be of such material and construction to facilitate thorough cleaning and to ensure that their use will not cause the adulteration of product during processing, handling, or storage. Equipment and utensils must be maintained in sanitary condition so as not to adulterate product.
b. Equipment and utensils must not be constructed, located, or operated in a manner that prevents FSIS personnel from inspecting the equipment or utensils to determine whether they are in sanitary condition.

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c. Receptacles used for storing inedible material must be of such material and construction that their use will not result in the adulteration of any edible product or in the creation of insanitary conditions. Such receptacles must not be used for storing any edible product and must bear conspicuous and distinctive marking to identify permitted uses.

Comments

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Food Code

Multiuse

4-101.11 Characteristics.

Materials that are used in the construction of utensils and food-contact surfaces of equipment may not allow the migration of deleterious substances or impart colors, odors, or tastes to food and under normal use conditions shall be:

A. Safe;
B. Durable, corrosion-resistant, and nonabsorbent;
C. Sufficient in weight and thickness to withstand repeated warewashing;
D. Finished to have a smooth, easily cleanable surface; and
E. Resistant to pitting, chipping, crazing, scratching, scoring, distortion, and decomposition.

4-101.16 Sponges, Use Limitation.

Sponges may not be used in contact with cleaned and sanitized or in-use food-contact surfaces.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

4-101.18 Lead in Solder and Flux, Use Limitation.

Solder and flux containing lead in excess of 0.2% may not be used as a food-contact surface.

4-101.19 Wood, Use Limitation.

A. Except as specified in (B), (C), and (D) of this section, wood and wood wicker may not be used as a food-contact surface.
B. Hard maple or an equivalently hard, close-grained wood may be used for:
    1. Cutting boards; cutting blocks; bakers' tables; and utensils such as rolling pins, doughnut dowels, salad bowls, and chopsticks; and

4-101.111 Nonfood-Contact Surfaces.

Nonfood-contact surfaces of equipment that are exposed to splash, spillage, or other food soiling or that require frequent cleaning shall be constructed of a corrosion-resistant, nonabsorbent, and smooth material.

Single-Service and Single-Use

4-102.11 Characteristics.

Materials that are used to make single-service and single-use articles:

A. May not:
    1. Allow the migration of deleterious substances, or
B. Impart colors, odors, or tastes to food; and

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

C. Shall be:

1. Safe, and
2. Clean

Durability and Strength

4-201.11 Equipment and Utensils.

Equipment and utensils shall be designed and constructed to be durable and to retain their characteristic qualities under normal use conditions.

4-201.12 Food Temperature Measuring Devices.

Food temperature measuring devices may not have sensors or stems constructed of glass, except that thermometers with glass sensors or stems that are encased in a shatterproof coating such as candy thermometers may be used.

Cleanability

4-202.11 Food-Contact Surfaces.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

A. Multiuse food-contact surfaces shall be:

1. Smooth;
2. Free of breaks, open seams, cracks, chips, inclusions, pits, and similar imperfections;
3. Free of sharp internal angles, corners, and crevices;
4. Finished to have smooth welds and joints; and
5. Except as specified in (B) of this section, accessible for cleaning and inspection by one of the following methods:
    a. Without being disassembled,
    b. By disassembling without the use of tools, or
    c. By easy disassembling with the use of handheld tools commonly available to maintenance and cleaning personnel such as screwdrivers, pliers, open-end wrenches, and Allen wrenches.

B. Subparagraph (A)(5) of this section does not apply to cooking oil storage tanks, distribution lines for cooking oils, or beverage syrup lines or tubes

4-202.12 CIP (Clean-in Place) Equipment.

A. CIP equipment shall meet the characteristics specified under 4-202.11 and shall be designed and constructed so that:

1. Cleaning and sanitizing solutions circulate throughout a fixed system and contact all interior food-contact surfaces, and
2. The system is self-draining or capable of being completely drained of cleaning and sanitizing solutions; and

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B. CIP equipment that is not designed to be disassembled for cleaning shall be designed with inspection access points to ensure that all interior food-contact surfaces throughout the fixed system are being effectively cleaned.

4-202.16 Nonfood-Contact Surfaces.

Nonfood-contact surfaces shall be free of unnecessary ledges, projections, and crevices, and designed and constructed to allow easy cleaning and to facilitate maintenance.

4-202.18 Ventilation Hood Systems, Filters.

Filters or other grease extracting equipment shall be designed to be readily removable for cleaning and replacement if not designed to be cleaned in place.

Functionality

4-204.11 Ventilation Hood Systems, Drip Prevention.

Exhaust ventilation hood systems in food preparation and warewashing areas including components such as hoods, fans, guards, and ducting shall be designed to prevent grease or condensation from draining or dripping onto food, equipment, utensils, linens, and single-service and single-use articles.

4-204.12 Equipment Openings, Closures and Deflectors.

A. A cover or lid for equipment shall overlap the opening and be sloped to drain.
B. An opening located within the top of a unit of equipment that is designed for use with a cover or lid shall be flanged upward at least 5 millimeters (two-tenths of an inch).

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

C. Except as specified under (D) of this section, fixed piping, temperature measuring devices, rotary shafts, and other parts extending into equipment shall be provided with a watertight joint at the point where the item enters the equipment.

D. If a watertight joint is not provided:

1. The piping, temperature measuring devices, rotary shafts, and other parts extending through the openings shall be equipped with an apron designed to deflect condensation, drips, and dust from openings into the food; and
2. The opening shall be flanged as specified under (B) of this section.

4-204.15 Bearings and Gear Boxes, Leakproof.

Equipment containing bearings and gears that require lubricants shall be designed and constructed so that the lubricant can not leak, drip, or be forced into food or onto food-contact surfaces.

4-204.18 Condenser Unit, Separation.

If a condenser unit is an integral component of equipment, the condenser unit shall be separated from the food and food storage space by a dustproof barrier.

4-204.120 Equipment Compartments, Drainage.

Equipment compartments that are subject to accumulation of moisture due to conditions such as condensation, food or beverage drip, or water from melting ice shall be sloped to an outlet that allows complete draining.

Acceptability

4-205.10 Food Equipment, Certification and Classification.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

Food equipment that is certified or classified for sanitation by an American National Standards Institute (ANSI)-accredited certification program will be deemed to comply with Parts 4-1 and 4-2 of this chapter.

Location

4-401.11 Equipment, Clothes Washers and Dryers, and Storage Cabinets, Contamination Prevention.

A. Except as specified in (B) of this section, equipment, a cabinet used for the storage of food, or a cabinet that is used to store cleaned and sanitized equipment, utensils, laundered linens, and single-service and single-use articles may not be located:
    1. In locker rooms;
    2. In toilet rooms;
    3. In garbage rooms;
    4. In mechanical rooms;
    5. Under sewer lines that are not shielded to intercept potential drips;
    6. Under leaking water lines including leaking automatic fire sprinkler heads or under lines on which water has condensed;
    7. Under open stairwells; or
    8. Under other sources of contamination.

Installation

4-402.11 Fixed Equipment, Spacing or Sealing.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

A. Equipment that is fixed because it is not easily movable shall be installed so that it is:

1. Spaced to allow access for cleaning along the sides, behind, and above the equipment;
2. Spaced from adjoining equipment, walls, and ceilings a distance of not more than 1 millimeter or one thirty-second inch; or
3. Sealed to adjoining equipment or walls, if the equipment is exposed to spillage or seepage.

B. Table-mounted equipment that is not easily movable shall be installed to allow cleaning of the equipment and areas underneath and around the equipment by being:

1. Sealed to the table; or
2. Elevated on legs as specified under 4-402.12(D).

4-402.12 Fixed Equipment, Elevation or Sealing.

A. Except as specified in (B) and (C) of this section, floor-mounted equipment that is not easily movable shall be sealed to the floor or elevated on legs that provide at least a 15 centimeter (6 inch) clearance between the floor and the equipment.

B. If no part of the floor under the floor-mounted equipment is more than 15 centimeters (6 inches) from the point of cleaning access, the clearance space may be only 10 centimeters (4 inches).

C. Except as specified in (E) of this section, table-mounted equipment that is not easily movable shall be elevated on legs that provide at least a 10 centimeter (4 inch) clearance between the table and the equipment.

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D. The clearance space between the table and table-mounted equipment may be:

1. 7.5 centimeters (3 inches) if the horizontal distance of the table top under the equipment is no more than 50 centimeters (20 inches) from the point of access for cleaning; or
2. 5 centimeters (2 inches) if the horizontal distance of the table top under the equipment is no more than 7.5 centimeters (3 inches) from the point of access for cleaning.

Equipment

4-501.11 Good Repair and Proper Adjustment.

A. Equipment shall be maintained in a state of repair and condition that meets the requirements specified under Parts 4-1 and 4-2.

B. Equipment components such as doors, seals, hinges, fasteners, and kick plates shall be kept intact, tight, and adjusted in accordance with manufacturer's specifications.

C. Cutting or piercing parts of can openers shall be kept sharp to minimize the creation of metal fragments that can contaminate food when the container is opened.

4-501.12 Cutting Surfaces.

Surfaces such as cutting blocks and boards that are subject to scratching and scoring shall be resurfaced if they can no longer be effectively cleaned and sanitized, or discarded if they are not capable of being resurfaced.

Other Sources of Guidance

§ 416.4 Sanitary operations.

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USDA may not cite, use, or rely on any guidance that is not available through their guidance portal, except to establish historical facts.

a. All food-contact surfaces, including food-contact surfaces of utensils and equipment, must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions or the adulteration of product.
b. Non-food-contact surfaces of facilities, equipment, and utensils used in the operation of the establishment must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions or the adulteration of product.
c. Cleaning compounds, sanitizing agents, processing aids, and other chemicals used by an establishment must be safe and effective under the conditions of use. Such chemicals must used, handled, and stored in a manner that will not adulterate product or create insanitary conditions. Documentation substantiating the safety of a chemical's use in a food processing environment must be available to FSIS inspection personnel for review.
d. Product must be protected from adulteration during processing, handling, storage, loading, and unloading at and during transportation from official establishments.

Comments

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Food Code

Objective

4-601.11 Equipment, Food-Contact Surfaces, Nonfood-Contact Surfaces, and Utensils.*

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Equipment food-contact surfaces and utensils shall be clean to sight and touch.
The food-contact surfaces of cooking equipment and pans shall be kept free of encrusted grease deposits and other soil accumulations.
Nonfood-contact surfaces of equipment shall be kept free of an accumulation of dust, dirt, food residue, and other debris.

Frequency

4-602.11 Equipment Food-Contact Surfaces and Utensils.

4-602.12 Cooking and Baking Equipment.

The food-contact surfaces of cooking and baking equipment shall be cleaned at least every 24 hours. This section does not apply to hot oil cooking and filtering equipment if it is cleaned as specified in Subparagraph 4-602.11(D)(6).

4-602.13 Nonfood-Contact Surfaces.

Nonfood-contact surfaces of equipment shall be cleaned at a frequency necessary to preclude accumulation of soil residues.

Methods

4-603.11 Dry Cleaning.

4-603.12 Precleaning.

4-603.14 Wet Cleaning.

4-701.10 Food-Contact Surfaces and Utensils.

4-702.11 Before Use After Cleaning.

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4-703.11 Hot Water and Chemical.

After being cleaned, equipment food-contact surfaces and utensils shall be sanitized in:

Drying

4-901.11 Equipment and Utensils, Air-Drying Required.

After cleaning and sanitizing, equipment and utensils:

4-901.12 Wiping Cloths, Air-Drying Locations.

Wiping cloths laundered in a food establishment that does not have a mechanical clothes dryer as specified in § 4-301.15(B) shall be air-dried in a location and in a manner that prevents contamination of food, equipment, utensils, linens, and single-service and single-use articles and the wiping cloths. This section does not apply if wiping cloths are stored after laundering in a sanitizing solution as specified under § 4-501.114.

Lubricating and Reassembling

4-902.11 Food-Contact Surfaces.

Lubricants shall be applied to food-contact surfaces that require lubrication in a manner that does not contaminate food-contact surfaces.

4-902.12 Equipment.

Equipment shall be reassembled so that food-contact surfaces are not contaminated.

Storing

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4-903.11 Equipment, Utensils, Linens, and Single-Service and Single-Use Articles.

(A) Except as specified in (D) of this section, cleaned equipment and utensils, laundered linens, and single-service and single-use articles shall be stored:

(B) Clean equipment and utensils shall be stored as specified under (A) of this section and shall be stored:

(C) Single-service and single-use articles shall be stored as specified under (A) of this section and shall be kept in the original protective package or stored by using other means that afford protection from contamination until used.

(D) Items that are kept in closed packages may be stored less than 15 cm (6 inches) above the floor on dollies, pallets, racks, and skids that are designed as specified under § 4-204.122.

4-903.12 Prohibitions.

Except as specified in (B) of this section, cleaned and sanitized equipment, utensils, laundered linens, and single-service and single-use articles may not be stored:

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Other Sources of Guidance

- (A) Equipment food-contact surfaces and utensils shall be cleaned:
  1. Except as specified in (B) of this section, before each use with a different type of raw animal food such as beef, fish, lamb, pork, or poultry;
  2. Each time there is a change from working with raw foods to working with ready-to-eat foods;
  3. Between uses with raw fruits and vegetables and with potentially hazardous food;
  4. Before using or storing a food temperature measuring device; and
  5. At any time during the operation when contamination may have occurred.
- (B) Subparagraph (A)(1) of this section does not apply if the food-contact surface or utensil is in contact with a succession of different raw animal foods each requiring a higher cooking temperature as specified under 3-401.11 than the previous food, such as preparing raw fish followed by cutting raw poultry on the same cutting board.
- Except as specified in (D) of this section, if used with potentially hazardous food, equipment food-contact surfaces and utensils shall be cleaned throughout the day at least every 4 hours.

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- Surfaces of utensils and equipment contacting potentially hazardous food may be cleaned less frequently than every 4 hours if:
    1. In storage, containers of potentially hazardous food and their contents are maintained at temperatures specified under Chapter 3 and the containers are cleaned when they are empty;
    2. Utensils and equipment are used to prepare food in a refrigerated room or area that is maintained at one of the temperatures in the following chart and:
        - The utensils and equipment are cleaned at the frequency in the following chart that corresponds to the temperature:

Temperature	Cleaning Frequency
5.0°C (41°F) or less	24 hours
>5.0°C - 7.2°C (>41°F - 45°F)	20 hours
>7.2°C - 10.0°C (>45°F - 50°F)	16 hours
>10.0°C - 12.8°C (>50°F - 55°F)	10 hours

    3. The cleaning frequency based on the ambient temperature of the refrigerated room or area is documented in the food establishment.
    4. Temperature measuring devices are maintained in contact with food, such as when left in a container of deli food or in a roast, held at temperatures specified under Chapter 3;
    5. Equipment is used for storage of packaged or unpackaged food such as a reach-in refrigerator and the equipment is cleaned at a frequency necessary to preclude accumulation of soil residues;

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6. The cleaning schedule is approved based on consideration of:
    - Characteristics of the equipment and its use,
    - The type of food involved,
    - The amount of food residue accumulation, and
    - The temperature at which the food is maintained during the operation and the potential for the rapid and progressive multiplication of pathogenic or toxigenic microorganisms that are capable of causing foodborne disease,
    - or, in-use utensils are intermittently stored in a container of water in which the water is maintained at 60°C (140°F) or more and the utensils and container are cleaned at least every 24 hours or at a frequency necessary to preclude accumulation of soil residues.

- Except when dry cleaning methods are used as specified under 4-603.11, surfaces of utensils and equipment contacting food that is not potentially hazardous shall be cleaned:
    1. At any time when contamination may have occurred;
- If used, dry cleaning methods such as brushing, scraping, and vacuuming shall contact only surfaces that are soiled with dry food residues that are not potentially hazardous.
- Cleaning equipment used in dry cleaning food-contact surfaces may not be used for any other purpose.
- Food debris on equipment and utensils shall be scrapped over a waste disposal unit, scupper, or garbage receptacle or shall be removed in a warewashing machine with a prewash cycle.
- If necessary for effective cleaning, utensils and equipment shall be preflushed, presoaked, or scrubbed with abrasives.

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- Equipment food-contact surfaces and utensils shall be effectively washed to remove or completely loosen soils by using the manual or mechanical means necessary such as the application of detergents containing wetting agents and emulsifiers; acid, alkaline, or abrasive cleaners; hot water; brushes; scouring pads; high-pressure sprays; or ultrasonic devices.
- The washing procedures selected shall be based on the type and purpose of the equipment or utensil, and on the type of soil to be removed.
- Equipment food-contact surfaces and utensils shall be sanitized.
- Utensils and food-contact surfaces of equipment shall be sanitized before use after cleaning.
- (A) Hot water manual operations by immersion for at least 30 seconds and as specified under § 4-501.111;
- (B) Hot water mechanical operations by being cycled through equipment that is set up as specified under §§ 4-501.15, 4-501.112, and 4-501.113 and achieving a utensil surface temperature of 71°C (160°F) as measured by an irreversible registering temperature indicator; or

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- (C) Chemical manual or mechanical operations, including the application of sanitizing chemicals by immersion, manual swabbing, brushing, or pressure spraying methods, using a solution as specified under § 4-501.114 by providing:
    1. Except as specified under Subparagraph (C)(2) of this section, an exposure time of at least 10 seconds for a chlorine solution specified under § 4-501.114(A),
    2. An exposure time of at least 7 seconds for a chlorine solution of 50 mg/L that has a pH of 10 or less and a temperature of at least 38°C (100°F) or a pH of 8 or less and a temperature of at least 24°C (75°F),
    3. An exposure time of at least 30 seconds for other chemical sanitizing solutions, or
    4. An exposure time used in relationship with a combination of temperature, concentration, and pH that, when evaluated for efficacy, yields sanitization as defined in Subparagraph 1-201.10(B)(70).
- Shall be air-dried or used after adequate draining as specified in § (a) of 21 CFR 178.1010 Sanitizing solutions, before contact with food; and
- May not be cloth dried except that utensils that have been air-dried may be polished with cloths that are maintained clean and dry.

1. In a clean, dry location;
2. Where they are not exposed to splash, dust, or other contamination; and
3. At least 15 cm (6 inches) above the floor.
4. In a self-draining position that allows air drying; and
5. Covered or inverted.

- In locker rooms;

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- In toilet rooms;
- In garbage rooms;
- In mechanical rooms;
- Under sewer lines that are not shielded to intercept potential drips;
- Under leaking water lines including leaking automatic fire sprinkler heads or under lines on which water has condensed;
- Under open stairwells; or
- Under other sources of contamination.

§ 416.5 Employee Hygiene.

a. Cleanliness. All persons working in contact with product, food-contact surfaces, and product-packaging materials must adhere to hygienic practices while on duty to prevent adulteration of product.
b. Clothing. Aprons, frocks, and other outer clothing worn by persons who handle product must be of material that is disposable or readily cleaned. Clean garments must be worn at the start of each working day and garments must be changed during the day as often as necessary to prevent contamination or adulteration of product.
c. Disease control. Any person who has or appears to have an infectious disease, open lesion, including boils, sores, or infected wounds, or any other abnormal source of microbial contamination must be excluded from any operations which could result in product adulteration until the condition is corrected.

Comments

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Food Code

2-301.11 Clean Condition.

Food employees shall keep their hands and exposed portions of their arms clean.

2-301.12 Cleaning Procedure.

A. Except as specified in (B) of this section, food employees shall clean their hands and exposed portions of their arms with a cleaning compound in a lavatory that is equipped as specified under § 5-202.12 by vigorously rubbing together the surfaces of their lathered hands and arms for at least 20 seconds and thoroughly rinsing with clean water. Employees shall pay particular attention to the areas underneath the fingernails and between the fingers.
B. If approved and capable of removing the types of soils encountered in the food operations involved, an automatic handwashing facility may be used by food employees to clean their hands.

2-301.14 When to Wash.

Food employees shall clean their hands and exposed portions of their arms as specified under § 2-301.12 immediately before engaging in food preparation including working with exposed food, clean equipment and utensils, and unwrapped single-service and single-use articles and:

A. After touching bare human body parts other than clean hands and clean, exposed portions of arms;
B. After using the toilet room;
C. After caring for or handling service animals or aquatic animals as specified in § 2-403.11(B);

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D. Except as specified in §2-401.11(B), after coughing, sneezing, using a handkerchief or disposable tissue, using tobacco, eating, or drinking;

E. After handling soiled equipment or utensils;

F. During food preparation, as often as necessary to remove soil and contamination and to prevent cross contamination when changing tasks;

G. When switching between working with raw food and working with ready-to-eat food; and

H. After engaging in other activities that contaminate the hands.

2-301.15 Where to Wash.

Food employees shall clean their hands in a handwashing lavatory or approved automatic handwashing facility and may not clean their hands in a sink used for food preparation, or in a service sink or a curbed cleaning facility used for the disposal of mop water and similar liquid waste.

2-301.16 Hand Sanitizers.

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A. A hand sanitizer and a chemical hand sanitizing solution used as a hand dip shall:

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1. Comply with one of the following:
    a. Be an approved drug that is listed in the FDA publication Approved Drug Products with Therapeutic Equivalence Evaluations as an approved drug based on safety and effectiveness; or
    b. Have active antimicrobial ingredients that are listed in:
        i. The FDA monograph for OTC Health-Care Antiseptic Drug Products as an antiseptic handwash, or
        ii. and (2) Comply with one of the following:
            a. Have components that are exempted from the requirement of being listed in federal food additive regulations as specified in 21 CFR 170.39 - Threshold of regulation for substances used in food-contact articles; or
            b.
            c. Comply with and be listed in:
                i. 21 CFR 178 - Indirect Food Additives: Adjuvants, Production Aids, and Sanitizers as regulated for use as a food additive with conditions of safe use, or
                ii.
                iii. 21 CFR 182 - Substances Generally Recognized as Safe, 21 CFR 184 Direct Food Substances Affirmed as Generally Recognized as Safe, or 21 CFR 186 - Indirect Food Substances Affirmed as Generally Recognized as Safe for use in contact with food; and
                iv. Be applied only to hands that are cleaned as specified under § 2-301.12

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B. If a hand sanitizer or a chemical hand sanitizing solution used as a hand dip does not meet the criteria specified under Subparagraph (A)(2) of this section, use shall be:

1. Followed by thorough hand rinsing in clean water before hand contact with food or by the use of gloves; or
2. Limited to situations that involve no direct contact with food by the bare hands.

C. A chemical hand sanitizing solution used as a hand dip shall be maintained clean and at a strength equivalent to at least 100 mg/L chlorine.

Fingernails

2-302.11 Maintenance.

A. Food employees shall keep their fingernails trimmed, filed, and maintained so the edges and surfaces are cleanable and not rough.

B. Unless wearing intact gloves in good repair, a food employee may not wear fingernail polish or artificial fingernails when working with exposed food.

Jewelry

2-303.11 Prohibition.

While preparing food, food employees may not wear jewelry on their arms and hands. This section does not apply to a plain ring such as a wedding band.

Outer Clothing

2-304.11 Clean Condition.

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Food employees shall wear clean outer clothing to prevent contamination of food, equipment, utensils, linens, and single-service and single-use articles.

Food Contamination Prevention

2-401.11 Eating, Drinking, or Using Tobacco.

A. Except as specified in (B) of this section, an employee shall eat, drink, or use any form of tobacco only in designated areas where the contamination of exposed food; clean equipment, utensils, and linens; unwrapped single-service and single-use articles; or other items needing protection can not result.
B. A food employee may drink from a closed beverage container if the container is handled to prevent contamination of:
    1. The employee's hands;
    2. The container; and
    3. Exposed food; clean equipment, utensils, and linens; and unwrapped single-service and single-use articles.

2-401.12 Discharges from the Eyes, Nose, and Mouth.

Food employees experiencing persistent sneezing, coughing, or a runny nose that causes discharges from the eyes, nose, or mouth may not work with exposed food; clean equipment, utensils, and linens; or unwrapped single-service or single-use articles.

Hair Restraints

2-402.11 Effectiveness.

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A. Except as provided in (B) of this section, food employees shall wear hair restraints such as hats, hair coverings or nets, beard restraints, and clothing that covers body hair, that are designed and worn to effectively keep their hair from contacting exposed food; clean equipment, utensils, and linens; and unwrapped single-service and single-use articles.

Other Sources of Guidance

§ 416.6 Tagging insanitary equipment, utensils, rooms or compartments.

When a Program employee finds that any equipment, utensil, room, or compartment at an official establishment is insanitary or that its use could cause the adulteration of product, he will attach to it a "U.S. Rejected" tag. Equipment, utensils, rooms, or compartments so tagged cannot be used until made acceptable. Only a Program employee may remove a "U.S. Rejected" tag.

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